

Generation Interconnection System Impact Study Report

For Queue Project Q11-0009

Cholame 138 kV

August 21, 2012

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q11-0009. The Interconnection Customer proposes a 196.3 MW solar photovoltaic facility interconnecting at the Cholame 138 kV substation in the Virginia area. The request was received on February 08, 2011. The study evaluated system intact and single contingency conditions using the regional planning case, including 6 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 4 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$3,735,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

1. Facility: Wheeler – Firebantz Substation 138.0 kV
Contingency: Chaffee Gas Recovery – S Jordan 138.0 kV
Loading: 106.2% (Rating: 258 MVA)
DFAX: 14.6%
Pre-project loading: 95.1%
2. Facility: Clifton Substation – Colonial 138.0 kV Line 1
Contingency: Clifton – Colonial Substation 138.0 kV Line 2
Loading: 103.0% (Rating: 336 MVA)
DFAX: 6.4%
Pre-project loading: 99.3%
3. Facility: Sideburn – Colonial 138.0 kV
Contingency: Colonial – Idaho Springs 138.0 kV
Loading: 102.0% (Rating: 265 MVA)
DFAX: 5.0%
Pre-project loading: 98.3%
4. Facility: Colonial – Idaho Springs 138.0 kV L
Contingency: Sideburn – Colonial 138.0 kV line
Loading: 101.0% (Rating: 302 MVA)
DFAX: 6.4%
Pre-project loading: 96.8%

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Firebantz – Wheeler 138.0 kV	Reconductor line	\$1,434,000	100% (\$1,434,000)
Clifton – Colonial 138.0 kV line	Reconductor line	\$821,000	100% (\$821,000)
Sideburn – Colonial 138.0 kV	Reconductor line	\$974,000	100% (\$974,000)
Colonial – Idaho Springs 138.0 kV L	Reconductor line	\$506,000	100% (\$506,000)
	Total allocated to project		\$3,735,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.